Response to Professor Yong: Tool Design Analysis

Where Would the CCO Have Seen a Warning Before the Recession?

> *"Imagine a busy CCO receives your US report today and sees 'No major issues detected.' They approve the model. Six months later a recession hits, the ethnicity disparity ratio falls to 0.79 below the US threshold, the bank gets fined, and consumers are harmed. Where in your tool's current design would that CCO have seen a warning? What would the warning have looked like?"* — *Stanley Yong*

1. Where in the Current Design Would the CCO Have Seen a Warning?

Not in the compliance report. The US compliance report (compliance_reports.json) returns an empty findings list and the status "No major issues detected." The function run_compliance_check() evaluates only the current portfolio; the ethnicity disparity ratio of 0.858 exceeds the US threshold of 0.80, so no finding is raised.

The drift simulation does run in the same execution (run_demo.py line 43) and its output drift_comparison.csv shows the ethnicity disparity ratio falling to 0.799 under a recession shock — below 0.80. But run_compliance_check() never receives that file. A CCO reading only compliance_reports.json has no reason to look for the CSV.

The warning is visible in notebook cell 15 (which states explicitly that the ratio crosses the US threshold) and in slide 4 of the presentation. Neither is part of the compliance report.

2. What Would the Warning Have Looked Like?

The required change is to pass drift results into run_compliance_check() as an optional parameter (drift_df). If the stressed ratio falls below the threshold while the current ratio does not, a Medium-severity finding is emitted and the overall status escalates from "No major issues detected" to "Review Required."

Resulting entry in compliance_reports.json:

```
overall_status: "Review Required"
findings:
  - severity:  Medium
    rule_id:   STRESS-DISPARITY-PROXIMITY
    Current portfolio ratio:   0.858  (compliant, threshold 0.80)
    Recession scenario ratio:  0.799  (below threshold - breach)
    Headroom consumed:         0.058  (7.2% buffer, fully eroded)
    Action: internal alert at 0.83 for ~60-day lead time.
```

Why the Gap Exists — and the Regulatory Principle It Illustrates

3. Why the Gap Exists

The prototype deliberately separates drift simulation from compliance checking so that each module in src/ is independently testable. This gap is a known limitation, acknowledged in Section 6 of the Task 3 PDF.

In production, stress scenarios would be a required input to every compliance report, not an optional ad hoc analysis. The one-line fix — passing drift_df into run_compliance_check() — closes the loop. A jurisdiction-specific headroom threshold (e.g., US internal alert at 0.83, above the regulatory floor of 0.80) then converts the buffer into a structured early-warning mechanism with defined remediation lead time.

4. The Regulatory Principle

The scenario Professor Yong describes is precisely what OCC Bulletin 2026-13 addresses through stress-scenario model risk governance: a point-in-time compliance report is not evidence of ongoing compliance. The headroom metric (0.858 minus 0.80 = 0.058) operationalises this — compliance is a buffer, not a binary state.

The UK configuration partially addresses this. The flag requires_consumer_duty_outcome_monitoring is True in config_rules.py, and it already appears as an Info-level finding in every UK report, reflecting FCA Consumer Duty PS22/9's ongoing outcome-monitoring obligation. The US ECOA regime has no equivalent forward-looking stress requirement — this is a real regulatory asymmetry, not a gap in the prototype.

5. Summary

	Current prototype	Production requirement
CCO sees warning?	No (findings = [])	Yes - status: Review Required
Warning location?	drift_comparison.csv only	Embedded in compliance report
Severity level?	Not a finding	Medium severity
Rule ID?	N/A	STRESS-DISPARITY-PROXIMITY
Ratios shown?	0.799 in CSV only	0.858 baseline / 0.799 shocked
Headroom metric?	Not quantified	0.058; internal alert at 0.83
Regulatory basis?	Not referenced	OCC Bulletin 2026-13

In short: the tool correctly predicts the scenario Professor Yong describes, but stores the prediction in a secondary file rather than in the primary compliance report. Passing drift_df into run_compliance_check() would have changed the CCO's decision.